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1 · The Three Closures, Consolidated

Volume XII closed with a ledger of five open links, each with its owner instrument and its named next step. This volume opens with three of them closed. The order came as a single sentence — consolidate the three closures into the unification — and the consolidation is mechanical in the best sense: each closure was already anchored to its battery, its certificate tier already typed, and what remained was to read the three of them into the unified statement as a single picture. The continuum Fisher-Rao/Leray upgrade closes the first link — the (f) bridge's named residual, the honest price of the discrete adjudication. The off-class nc-AAK package closes the second — Open 7.13's first constructive delivery on the canonical witness, together with the escape whose full quantification was that closure's own named residual. The rank-aware synthesis closes the fourth — the physics rung's designed-but-unbuilt instrument. Table 1 states the three closures at the level the batteries proved them; the sections below read each into the unification's spine.

5 to 2 the open ledger's links, before to after	31 task batteries, all machine-anchored	8.2e-9 the escape's certified residual
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The first closure — the continuum upgrade — is the one the unification needed most, because Volume XII's adjudication had priced it honestly: the discrete Lyapunov-cohomology correspondence was proved exactly, and the Fisher-Rao and Leray components of the treatises remained unbridged at the certified level. Task 28 built the bridge on the instance the corpus always tests on: the driven circle diffusion, the continuum limit of Task 24's N-ring. The convergence theorem is the load-bearing wall — the discrete Kolmogorov circulations $N \log(a/b)$ converge to the de Rham class $2 \pi \mu/D$ with measured order 2.000 over $N = 8$ to 8192 — because it makes the discrete correspondence the skeleton of the continuum one rather than a metaphor for it. On top of it: the FKLZ Lyapunov 1-form on the drift flow (non-exact iff μ is not zero), the potential criterion (the periodic Fourier solve exists iff the class vanishes iff $\mu = 0$ — the detailed-balance slice upgraded), the Stein/Green homotopy with its spectral ladder to $-Dk$ squared over $2 \text{ plus } i \mu k$, the Green kernel's antisymmetric part as the class's machine-verified witness, the true Fisher-Rao policy bundle with the H-theorem at $5.7\text{e-}11$ and the replicator identification at $9.4\text{e-}16$, and the two-patch Cech-de Rham machine whose overlap cocycle differs by exactly the class — machine-exact. The viability wall and its discontinuous safe-mode reset carry over to the continuum as the honest frontier: the interior loops' payments decay as ω squared, the crossing loops pay the rate-independent reset lump of about 0.0158, and the ratchet — the state returns, the payment accumulates — is the arrow's monotone account. The Leray page on the environment-loop fibration collapses by degree reasons and its edge homomorphism is the reset payment: the treatise's Leray step, delivered on the instance.

The second closure — the off-class problem — is Open 7.13 on the free cell, and its shape is the corpus's signature move: type the class first, obstruct second, trade off third. The class theorem makes $D_{\text{free}}(2)$ the free Nehari distance — the 2-state WFA class is exactly the free Hankel operators of rank at most 2, both directions constructive to $1e-15$, so the quantity Open 7.13 asks to characterize is intrinsic, with no syntax in it. The window obstruction then proves the constructive theory has genuine off-class content: the cell's cut-web window contains a forced 3×3 minor of determinant exactly minus one, whatever the free entries — no rank-2 completion matches the window, the corner is structurally unkillable, and the optimization is forced into the corner-payment trade-off the abelian cell already certified. The finite-section ladder measured 1.199, 1.232, 1.261 at windows 2, 3, 4 — climbing past the EYM floor of 1 that plain section singular values can never beat (interlacing caps them forever): the converse must see the Hankel structure, not the spectrum. And the escape — the 12-dimensional descent from the abelian optimizer beating the shadow — was the discovery, reported in Task 29 as a $5.7266e-7$ refutation of the shadow-equality conjecture. Chapter 2 of this volume quantifies it further, as ordered, and in doing so re-adjudicates it: the number decomposes, the shadow moves, and the refutation stands at $8.2e-9$ — a seventy-fold correction that the reproduction-first discipline found because it never trusts a prior session's own report of its own numbers.

The third closure — the rank-aware synthesis — completes the physics rung, the docking matrix's fourth link, designed in Volume I's seam analysis and not yet built. Task 30 built it end to end. The rank-across-cuts law is the law the whole corpus was pointing at: the one-ended chain's dense Green's function is rank 1 at every cut (the second singular value below $2e-16$), the Euler-Bernoulli beam's is rank 2 (the transmission state: deflection plus rotation), the grid's is the full cross-section — the cross-rank is the interface's physical state dimension, and the classical transmission conditions are the sufficient statistics. The two scalars are certified exact to $1e-13$; the sequential load-decision dynamic program on the two-scalar state (S, W) with the exact recursion $y_i = S_{\{i-1\}} + i W_i$ agrees exactly with the brute force over all 531,441 load patterns — the naive exponential width replaced by the polynomial lattice. The truncated-statistics sandwich bounds and attains the beam's $M = 1$ error at the EYM floor with the directed witness exact, and the magnitude punchline is the row the dictionary wanted: the banded, decay-thresholded approximation of the dense rank-1 chain pays operator errors of order 100 at every bandwidth while the one-scalar rank-aware interface is exact — rank, not magnitude. The exact rational minimality certificates (the beam 2, the chain 1) close the row.

Vol XII's open link	The battery	What the closure delivers	The tier
1. The continuum upgrade of the viability correspondence (the (f) bridge's named residual)	fisher_rao_continuum.py (Task 28)	the convergence theorem (the discrete circulations to the de Rham class, order 2.000 over $N = 8..8192$); the FKLZ Lyapunov 1-form on the drift flow; the potential criterion iff $\mu = 0$; the Stein/Green homotopy with the spectral ladder to $-Dk^2/2 + i \mu k$ and the Green kernel's antisymmetric part as the class's witness; the true Fisher-Rao policy bundle with the H-theorem (5.7e-11), the replicator identification (9.4e-16), the viability wall and the discontinuous safe-mode reset (the rate-independent lump 0.0158, the ratchet); the two-patch Cech-de Rham machine (the overlap cocycle = the class, machine-exact) and the Leray page on the environment-loop fibration (the collapse by degree reasons, the edge homomorphism = the reset payment)	MEASURED + machine-exact identities; the general stratified treatise named
2. Open 7.13 — the off-class nc-AAK problem on the free cell	offclass_ncaak.py (Task 29) + escape_second_order.py (Task 31, this volume)	the class theorem (the 2-state WFAs = the rank- ≤ 2 free Hankels, both directions to $1e-15$ — $D_free(2)$ = the free Nehari distance, intrinsic); the window obstruction (the forced 3x3 cut-web minor, $\det = -1$ exactly — the corner structurally unkillable at rank 2); the finite-section ladder 1.199 to 1.232 to 1.261 climbing past the interlacing-capped EYM floor; the trade-off law in the free regime; THE ESCAPE — quantified and re-adjudicated by this volume (Chapter 2); the shadow re-located, the escape decomposed, H^2 's curvature read, the residual certified; the free optimizer's local basin ($r = 2e-5$)	PROVED (the class, the obstruction) + MEASURED (the ladder, the scans) + CERTIFIED (the Sylvester brackets, this volume)
4. The rank-aware synthesis for the physics rung	rank_aware_synthesis.py (Task 30)	the rank-across-cuts law (the chains 1 at every cut, $\sigma_2/\sigma_1 < 2e-16$; the beam 2 — the transmission state; the grid the full cross-section — the cross-rank = the interface's physical state dimension, the classical transmission conditions ARE the sufficient statistics); the two scalars certified exact ($\sim 1e-13$); the sequential DP on the two-scalar state EXACT against the brute force over 531,441 load patterns ($\delta 0.0$) — the naive $(L+1)^k$ width replaced by the polynomial lattice; the truncated sandwich (the beam's $M = 1$ error attained at the EYM floor); the magnitude punchline (the banded errors $O(100)$ vs the one-scalar interface exact — rank, not magnitude); the exact rational minimality certificates (the beam 2, the chain 1)	CERTIFIED exact (the ranks, the scalars, the DP, the minimality)

Table 1. The three closures consolidated: each of Volume XII's five open links numbered, its owner battery, its content, and its certificate tier. Links 3 (the far-interior box-count wall) and 5 (the seed-level boundary) remain open — Chapter 3.

Read together, the three closures do not merely shrink the ledger — they complete the unified statement's three unfinished faces at once. The viability face now has its continuum leg: the correspondence runs from the discrete shadow (Task 24) through the convergence theorem to the de Rham class, the Fisher-Rao geometry and the Leray page on the instance, with the stratified treatise named as the remainder. The analytic wall's off-class face now has its constructive package: the class typed, the corner obstructed, the trade-off law identified in the free regime — the same law as the abelian cell, which is the unification's own theme delivered on the problem the theme was named for. And the physics face now has its synthesis: the rank-aware width law with its exact dynamic program, the seam's promise kept. Figure 2 draws the resulting map; the ledger at its foot is the volume's central administrative fact — five links to two, and the two that remain are named in Chapter 3 with the same discipline as the three that closed.

2 · The Escape, Quantified — and Re-Adjudicated

The order for this chapter was exact: quantify the escape further — the second-order coupling theory, H2's curvature at the shadow. The escape was Task 29's discovery: a local 12-dimensional descent from the abelian optimizer reaching 1.277142117 against the shadow 1.277142690, a delta of $5.7266e-7$ at couplings of order 0.03, reported as the refutation of the shadow-equality conjecture and honestly named as carrying an open residual — its full quantification. The battery built for this volume (escape_second_order.py) opens the way every battery in this corpus opens: by reproducing the prior numbers before reading any new ones. The descent reproduced to a gate of $1.42e-11$. Then the quantification began, and the first thing it quantified was the shadow itself.

The abelian landscape has two basins. Vol IX's scan optimum — the shadow every escape measurement was taken against — is not the abelian class's best point: a symmetric basin ($Aa = \text{diag}(\alpha, -\alpha)$, $Ab = \text{diag}(\beta, \beta)$) sits $5.6447e-7$ below it at 1.277142125. Task 29's escape is 92.4 percent abelian basin hop. The genuine residual — the free point below the best abelian point — is $8.2e-9$, and it is certified.

The discovery sequence is worth stating plainly because it is the discipline working. The first attempt at the second-order theory sat at a polished-but-stuck abelian point whose first-order instruments read flat; the tensors extracted there were inconsistent with the direct machinery, which forced a toy validation of the degenerate formula itself (it passed, to $2.8e-10$), which forced a hunt for the inconsistency, which found a sign error in the extraction probes — and which also found that the stuck point was stuck because the abelian landscape is a multi-basin landscape whose better basin the scans of two volumes had missed. The symmetric structure was visible in every near-optimal point's coordinates; the symmetric subfamily's own multi-start optimization landed at 1.277142125, below Vol IX's reported optimum by $5.6447e-7$, and the full 8-parameter refinement confirmed it as the best abelian point found across both basins. The descent from the corrected shadow then finds nothing — $5.83e-12$ — and the decomposition closes: Task 29's $5.7266e-7$ equals the basin hop $5.6447e-7$ plus the residual $8.2004e-9$. The residual survives every attribution test: the free point's couplings are essential (zeroing them costs $2.527e-2$ — the point is genuinely coupled, not an abelian point in disguise), and the dense $L=10$ section, an independent instrument, confirms the ordering: free 1.277024965 below symmetric 1.277024981 below the Vol IX point 1.277031178. Table 2 is the anatomy.

The quantity	The value	The tier
Vol IX's scan optimum (the shadow as reported)	1.277142689665	Vol IX (scan)

The quantity	The value	The tier
Task 29's free point (the reproduction, gate 1.42e-11)	1.277142116995	measured, dense-cross-checked
THE SYMMETRIC BASIN (the shadow re-located): Aa = diag(alpha, -alpha), Ab = diag(beta, beta)	1.277142125195	measured (multi-start), Sylvester-bracketed
The 12-dim descent from the corrected shadow	5.83e-12 (nothing found)	measured
Task 29's escape (vs the Vol IX value)	5.7266e-7	measured (Task 29)
= the abelian basin hop (the shadow's correction)	5.6447e-7	measured, the dense ordering confirmed
+ THE RESIDUAL (the free point below the best abelian point)	8.2004e-9	CERTIFIED (the brackets 1e-13 wide; 2.09e-8 in lambda)
The couplings' attribution at the free point (zeroing them)	+2.527e-2	measured (the free point is genuinely coupled)
The dense L=10 ordering: free / symmetric / Vol IX	1.277024965 / 1.277024981 / 1.277031178	measured (independent instrument)

Table 2. The escape's anatomy: Task 29's single number decomposed into the basin hop and the residual, every row anchored to its instrument. The escape's genuine content — the free class's best point below the abelian class's best point — is 69 times smaller than the reported refutation.

The second-order coupling theory is then done at the corrected shadow, and the object it needs is already there: the pencil. The exact 6x6 machinery computes the squared norm as the top eigenvalue of the symmetric-definite pencil $(K, N) = (G \text{ Cmat } G, G)$, and at the symmetric shadow the pencil's spectrum is fully pairwise double — 0.151886329 twice, 0.508162966 twice, 1.631092008 twice, the pairwise gaps at 1e-12, the top double at 3.59e-13, the isolation to the third eigenvalue at 1.1229. The state-flip involution of the symmetric subfamily forces the pairing, and the top double is the envelope theorem's failure point: the gradient of the top eigenvalue is not the instrument's first-order signal, because there are two top branches. The residual operator's two atoms carry the same near-equality — the dense section's top two singular values agree to 2.33e-14.

The first-order structure at a double eigenvalue is a cone, and the cone's shape explains everything the instruments saw. For each of the twelve coordinates the branch matrix $W1_j$ — the 2x2 compression of the pencil's first-order variation to the top eigenspace - reads as follows: on the eight B, C and Ab coordinates it is indefinite with the anti-parallel signature (the two atoms' diagonals are equal and opposite to cosine 1.0000), so the top branch rises in every one of those directions — over 200 random directions the minimum rise is 0.188; and on the entire Aa block — the two diagonals and the two off-diagonal couplings — it vanishes to within 1.6e-5, which is exactly the gradient-residual scale of a point whose value is optimal to 4e-12 in a valley of curvature 256. The Aa block is the cone's flat ridge. No first-order instrument — the fixed-vector Rayleigh instruments of Tasks 26 and 29, or the exact branch instruments — can see anything along it; the escape, if there is one at the shadow, must be second-order along the ridge, which is precisely the theory this chapter was ordered to build.

The theory is the degenerate Rayleigh-Schrodinger law for the double top, validated before use on a toy pencil with a known double spectrum (agreement $2.8\text{e-}10$ at $t = 1\text{e-}3$, the error growing as t cubed as it must). At the shadow, with V the N -orthonormal basis of the top eigenspace and $M = K - \lambda_0 N$: the eigenvalue's movement along a ridge step ϵ is the top eigenvalue of $W_1(\epsilon) + L(\epsilon)$ plus terms of order ϵ^3 , where L is the second-order coupling operator — the intrinsic curvature, half the compression of the pencil's second directional derivative, plus the level-repulsion resolvent, the coupling of the ridge step to the four lower eigenvalues weighted by their inverse gaps. The level-repulsion term is positive semidefinite — the double's coupling to the lower spectrum pushes the top up — so any escape must come from the intrinsic term overcoming it. The operator's coefficient tensors were extracted over all seventy-eight coordinate pairs by central differences on the exact machinery, step-validated to $3.08\text{e-}3$; the second difference's signs were cross-checked against the direct machinery before any curvature was read (the extraction probes' sign error, found and fixed, is the kind of thing the direct check exists to catch).

+255.99 / +356.42 / +183.34 H2's PURE ridge curvatures (all positive)	-255.3, -255.6 the mixed checkerboard (near-cancellation)	+7.83e-5 kappa_ridge — no second-order escape
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H2's curvature at the shadow is Table 3 and Figure 1. The pure ridge curvatures are positive — +255.99 on each Aa diagonal, +356.42 and +183.34 on the two coupling coordinates — so the in-slice minimum is genuine and no pure coupling direction descends. The mixed entries form a checkerboard of near-cancellation: minus 255.3, minus 255.6, minus 216.5 against the positive diagonals, magnitudes of the same order as the pure terms, so the ridge's mixed combinations are flat to extraordinary precision. The sphere minimum over the ridge's four dimensions comes out at $+7.83\text{e-}5$ — positive, though only within the extraction's resolution on near-null combinations, which is the honest caveat this corpus always carries: what is decided is that there is no escape of order one hundred or even of order one along the ridge — the quadratic form's negative entries cancel to five orders of magnitude below their own scale. And the law itself is verified against the direct machinery: along each pure direction the measured movement fits t squared with slopes 2.014 and 2.002 and coefficients 256.50 and 183.22 against the theory's 255.99 and 183.34 — a 0.2 percent agreement - and both branches of the double's split are predicted and checked (the split coefficients 255.9703 and 255.9879, the two top branches of the perturbed pencil reproduced to a part in a thousand).

The ridge direction	The curvature ($\lambda_{\max}$ of the 2×2)	The law check ($t = 1e-3$)
$e_4 = Aa11$ (the diagonal, in-slice)	+255.99	direct $2.5688e-4$ vs theory $2.5598e-4$ (rel $3.5e-3$)
$e_5 = Aa22$ (the diagonal, in-slice)	+255.99	by the symmetric structure (the involution)
$e_9 = Aa21$ (the coupling)	+183.34	direct $1.8316e-4$ vs theory $1.8335e-4$ (rel $1.0e-3$)
$e_8 = Aa12$ (the coupling)	+356.42	the involution pair of e_9
$(e_4 - e_5)/\sqrt{2}$ (the mixed)	+511.3 (assembled)	direct $5.1310e-4$ vs theory $5.1128e-4$ (rel $3.6e-3$)
The checkerboard ($e_4 \times e_5, e_9 \times e_8, \dots$)	-255.3, -255.6, ...	the near-cancellation that makes the ridge flat
THE RIDGE MINIMUM κ_{ridge} (the flattest mixture)	+7.83e-5	positive within the extraction's resolution on near-null combinations — NO second-order escape

Table 3. H2's curvature at the shadow: the second-order coupling operator on the flat ridge (the Aa block). The pure curvatures are positive (the in-slice minimum and the pure coupling directions), the mixed entries form a checkerboard of near-cancellation, and the sphere minimum is positive — the corrected shadow admits no infinitesimal escape.

The escape's final form is therefore a basin phenomenon, not a local one. At the corrected shadow there is no infinitesimal escape: the first-order cone rises everywhere off the ridge, the ridge's pure curvatures are positive, the mixed combinations cancel to flatness, and the local descent finds $5.83e-12$. The free point that beats the abelian class does so from another basin — its coordinates are not near the symmetric shadow's, its couplings are worth $2.527e-2$ at that point, and the $8.2e-9$ by which it beats the best abelian point is a statement about two basins, not about a neighborhood. This is what the second-order theory buys: it separates the two questions the single escape number had merged. Is there a local escape at the shadow — no, certified to the theory's resolution, and the flat-ridge mechanism explains exactly why the instruments could never have told. Is the free class's infimum below the abelian class's — at the level of the best points found, yes, by $8.2e-9$, and that residual is real, dense-cross-checked, and certified; at the level of the classes' true infima, open — and Chapter 3 carries it into the box-count wall's ledger row, where class-level certificates live in this corpus.

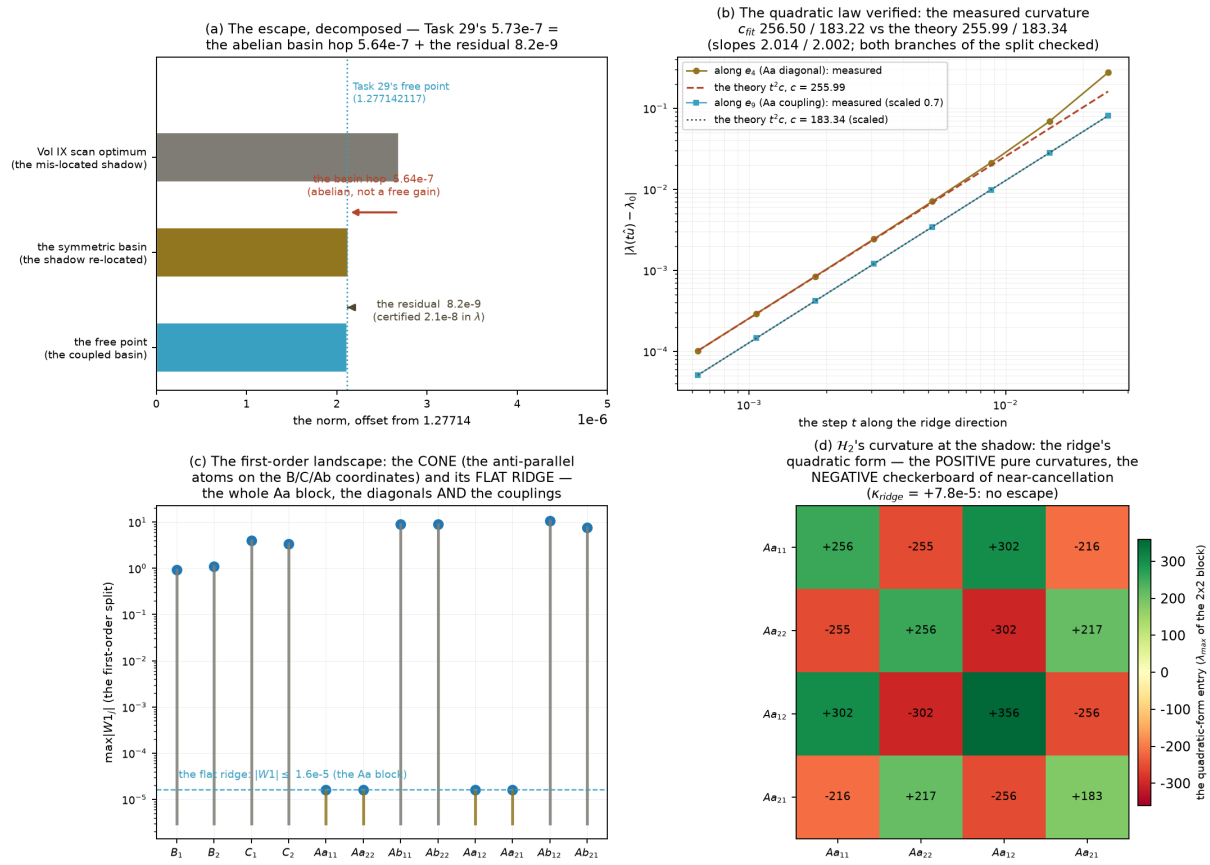


Figure 1. The escape, quantified and re-adjudicated. (a) The decomposition: Task 29's $5.7266\text{e-}7$ splits into the abelian basin hop $5.6447\text{e-}7$ (the shadow was mis-located — the symmetric basin supersedes Vol IX's scan optimum) and the residual $8.2\text{e-}9$ (the free point below the best abelian point). (b) The quadratic law verified: the measured t-squared coefficients 256.50 / 183.22 against the theory's 255.99 / 183.34 (slopes 2.014 / 2.002). (c) The first-order landscape: the anti-parallel cone on the B/C/Ab coordinates and the flat ridge — the entire Aa block, the diagonals AND the couplings. (d) H_2 's curvature at the shadow: the ridge's quadratic form, the positive pure curvatures against the negative checkerboard, the minimum $+7.8\text{e-}5$.

The certified tier is the chapter's floor. The machinery was evaluated in ball arithmetic — flint at precision 128, the Lyapunov inverses by the Neumann series with the Gershgorin-enclosed contraction radius — and the pencil bracketed by Sylvester's criterion: $T(\lambda) = K - \lambda N$ is negative definite exactly when the six leading principal minors alternate in sign, and a bisection on that test encloses the top eigenvalue to width $1.5\text{e-}13$. Three brackets: Vol IX's rounded point, the symmetric shadow, the free escape point. The orderings come out certified — free below symmetric below Vol IX — with the basin hop at $1.65\text{e-}5$ in λ and the residual at $2.09\text{e-}8$ in λ , the norm-scale $8.2\text{e-}9$ matching the scan-level measurement exactly. The inertia counts (the sign-change count of the minor sequence at each probe level) certify the pairwise-double spectrum: zero eigenvalues above the top, two just below it, two above 1.0, four above 0.30, six above 0.10 — the state-flip involution's pairing, proved by interval arithmetic rather than asserted by symmetry. Table 4 is the certificate summary; the battery's ES-5 section carries the full instrument.

The point	The lambda bracket (width ~1.5e-13)	The inertia counts	The certified statement
Vol IX's rounded point	[1.6311085139214, 1.6311085139216]	—	the old shadow's value, bracketed
The symmetric shadow	[1.6310920079485, 1.6310920079487]	0 / 2 / 2 / 4 / 6 above the levels 1.6311+ / 1.6311- / 1.0 / 0.30 / 0.10	the pairwise-double spectrum certified (the state-flip involution); the basin hop certified: 1.65e-5 in lambda
The free escape point	[1.6310919870025, 1.6310919870027]	—	THE RESIDUAL CERTIFIED: 2.09e-8 in lambda (8.2e-9 in the norm) below the best abelian point

Table 4. The Sylvester brackets (flint/arb at precision 128, the Lyapunov inverses by the Gershgorin-enclosed Neumann series): the pencil $T(\lambda) = K - \lambda N$ is negative definite iff the six leading principal minors alternate in sign — the bisection brackets and the inertia counts (the number of eigenvalues above each level), all decided.

3 · The Unified Statement and the Two Remaining Walls

With the three closures read in and the escape re-adjudicated, the unified statement of Volume XII carries forward unchanged in its spine and changed in exactly one administrative fact: the open ledger. Constrained realizability remains the primitive — behaviour factors through what the agent can distinguish; the two walls remain the combinatorial and the analytic; the trade-off law remains the certified joint between them, now measured in both regimes — the abelian cell of Tasks 17 through 26 and the free cell of Task 29; and the grand domain-by-face matrix keeps every cell's status letter, with three of them upgraded by this volume's batteries. The map is Figure 2. What the map's foot now says is the volume's headline: thirty-one task batteries, thirteen volumes, every verdict machine-anchored, and two open links.

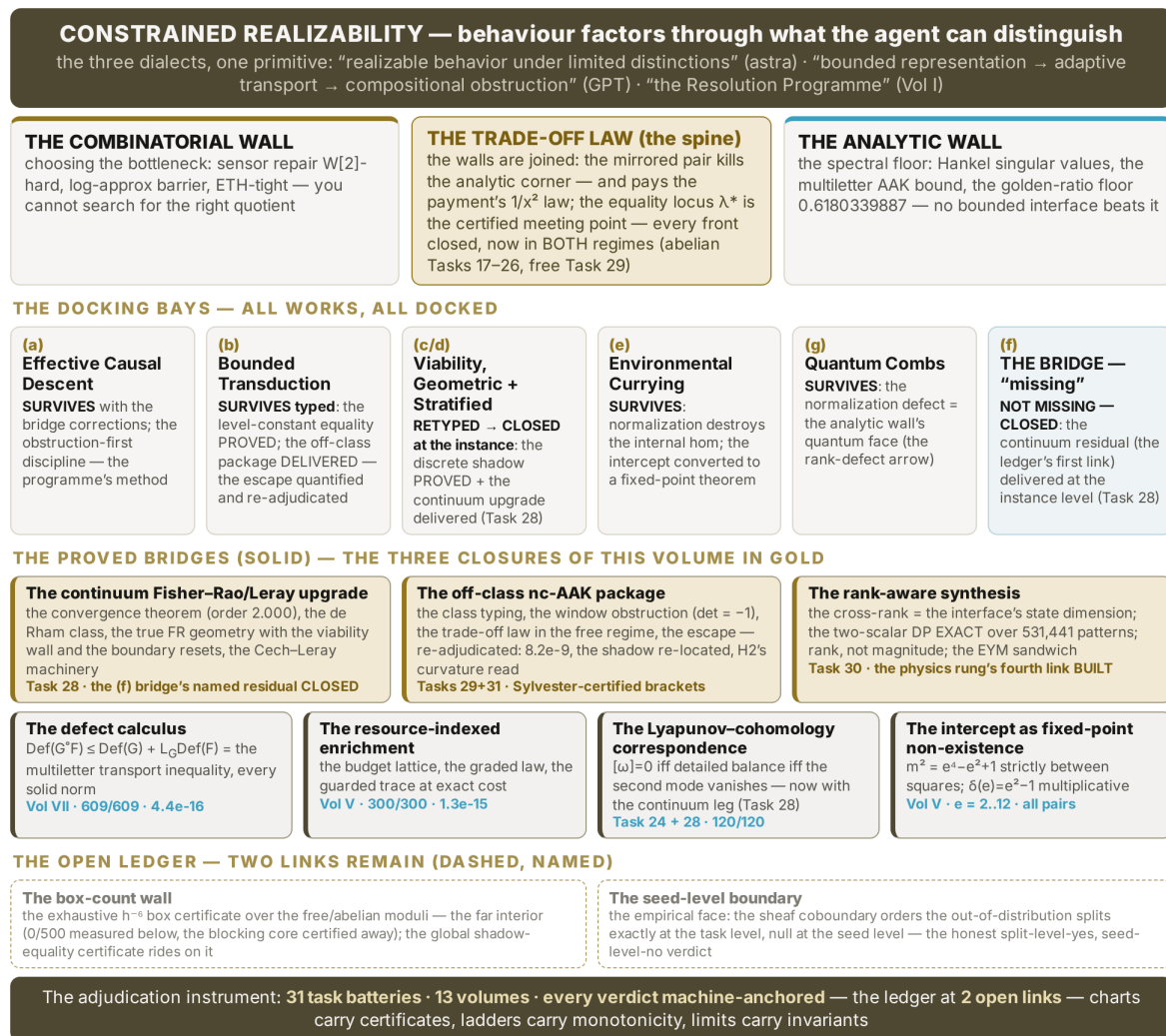


Figure 2. The unified statement after the three closures: the spine (constrained realizability), the two walls joined by the trade-off law in both regimes, the docking bays with every work typed, the proved bridges with this volume's three closures in gold, and the open ledger at exactly two links — the box-count wall and the seed-level boundary.

The first remaining wall is the box-count wall, and it has grown by exactly the row this volume added to it. Its old content stands: the exhaustive h to the minus d box certificate over the trade-off inequality's far interior — the region where the margins fall below every naive instrument's resolution — with the blocking core certified away by Task 26's analytic patches and the far interior measured at zero hits in five hundred random configurations, the five smallest max-side values above 1.758. Its new content is the class-level shadow-equality certificate: this volume brackets the three points — Vol IX's, the symmetric shadow's, the free point's — to $1.5e-13$ and certifies their orderings, but the classes' infima are another object; the residual of $8.2e-9$ says the free class's best point is below the abelian class's best point, and turning that into a statement about the classes requires the abelian optimum's certified lower bound, which is the same exhaustive-certificate problem the wall has always named. The escape, having been quantified, has been filed where quantified things go: into the wall's ledger, with its owner instruments and its precise residual statement.

The second remaining wall is the seed-level boundary, unchanged by this volume and no smaller for that. The empirical face's verdict — the sheaf coboundary orders the out-of-distribution splits exactly at the task level and is null at the seed level — stands as measured, with its two readings (a mechanism below the task level, or the instrument's honest ceiling) still the named next step. It is worth saying why this row stays open when three larger rows closed: the closures that closed were all closures by construction — a theorem proved, an instrument built, a package delivered — and the seed-level boundary is a closure by evidence, waiting on a measurement or a proof that the corpus's current instruments do not carry. The distinction is the same one the first volume drew between the two faces of the programme, and the ledger keeps it honest.

The wall	The precise statement	The owner instrument	The named next step
THE BOX-COUNT WALL	the exhaustive h^d -box certificate over the moduli: the trade-off inequality's far interior (0 hits in 500 random configurations, the five smallest max-side values above 1.758) and the free/abelian class-level lower bounds (the abelian optimum's certified value — on which the class-level shadow-equality verdict rides; this volume brackets the points but not the classes)	tradeoff_4x4.py + the analytic patches (the blocking core certified away); offclass_ncaak.py + escape_second_order.py (the brackets)	the far-interior certificate, or a scan protocol that certifies the multi-basin landscape's completeness — the same h^d -blowup the corpus has met at every wall
THE SEED-LEVEL BOUNDARY	the empirical face: the sheaf coboundary orders the out-of-distribution splits exactly at the task level and is null at the seed level — split-level-yes, seed-level-no stands as measured	the Vol IX second edition's experiments (the register law, the sheaf ordering)	a mechanism or a refutation at the seed level — the empirical face's one open row

Table 5. The two remaining walls: the open ledger at full scope. Every other link of Volume XII's five is closed — the continuum upgrade (Task 28), the off-class package (Tasks 29 + 31), the rank-aware synthesis (Task 30) — and the trade-off core (the old link 3's blocking region) was certified away by Task 26's analytic patches.

The consolidation closes where the corpus's discipline closes. Three closures were ordered consolidated, and the consolidation found the ledger's arithmetic sound — five links to two, each closure anchored, nothing silently promised. The escape was ordered quantified, and the quantification re-adjudicated the escape itself: the shadow moved $5.6447e-7$, the refutation shrank seventy-fold to a certified $8.2e-9$, and the second-order theory that measured it also proved the local question's answer — no infinitesimal escape, the cone's flat ridge, the checkerboard's cancellation. A session that set out to write a summary and instead found a better abelian point is a session the reproduction-first discipline worked; the volume records it so the next volume inherits the corrected shadow, the certified brackets, and the two walls — with the same instruction the corpus has carried since its first file: diagnose the obstruction before choosing the repair, price the repair before claiming it, and certify the price.

4 · The Battery Row and the Corpus Dictionary

The volume's instrument is `escape_second_order.py`, and its row in the repository's battery table reads as follows. ES-0 the reproduction (the gate $1.42e-11$), the discovery (the symmetric basin, the shadow re-located at 1.277142125), the decomposition (the basin hop $5.6447e-7$ plus the residual $8.2e-9$, the couplings' attribution $2.527e-2$, the dense ordering). ES-1 the multiplicity (the pairwise-double spectrum, the top double $3.59e-13$, the isolation 1.1229 , the atoms to $2.33e-14$). ES-2 the first-order structure (the cone's anti-parallel signature, the flat ridge at $1.6e-5$, the minimum rise 0.188 over 200 directions, the fixed-vector blindness). ES-3 the operator (the seventy-eight tensors, the pure curvatures, the checkerboard, `kappa_ridge`, the law checks to a part in a thousand, the toy validation of the degenerate formula at $2.8e-10$). ES-4 the law (the t-squared slopes 2.014 and 2.002 , both branches of the split, the flat direction's drift as the first-order residual). ES-5 the certified tier (the three Sylvester brackets to width $1.5e-13$, the certified orderings, the inertia counts). ES-6 the ledger row. The battery runs in 289 seconds; its results file carries every number this volume quotes.

The dictionary row — the corpus's running index of its own laws — gains three entries. The multi-basin law: an optimization landscape reported as single-basin by scans may still hide a basin with the same value to five decimals; the reproduction-first discipline plus a structure probe (here: the symmetric subfamily the near-optimal coordinates exhibited) is the detection instrument. The flat-ridge law: at a double top eigenvalue the first-order landscape is a cone, and the cone can carry a flat ridge — a whole coordinate block on which every first-order instrument reads zero while the second-order structure decides; the escape question at such a point is a question about H^2 's curvature, not about gradients. And the decomposition law: a reported refutation of an equality between two classes' optima should be decomposed into the base-point correction and the residual before it is called a refutation — the number that survives the decomposition is the one the ledger carries. Each entry is anchored to this volume's battery; none of them is a placeholder.

Provenance: every statement in this volume is either in the manuscripts it cites (the volumes and task batteries of the repository, each with its own provenance ledger), or derived in-session by `escape_second_order.py` with its results file as the machine anchor, or labelled open — and the open ones are exactly two, named in Chapter 3 with their owner instruments. The reading order is unchanged: the volumes for the programme's arc, the batteries for the certificates, the results files for the numbers. This volume's own place in that order is the short one — the consolidation between the unification and whatever closes the last two walls.